would be an ancestor/descendant relationship. A new blemish that showed up anywhere along the series would be shared by descendants but not by ancestors. An ancestor/descendant series of this kind has the potential to evolve.

Superficially, successive generations of stick-insect bodies appear to constitute a lineage of replicas. But if you experimentally change one member of the lineage (for instance by removing a leg), the change is not passed on down the lineage. By contrast, if you experimentally change one member of the lineage of genomes (for instance by X-rays), the change will be passed on down the lineage. This, rather than the fragmenting effect of meiosis, is the fundamental reason for saying that the individual organism is not the 'unit of selection'—not a true replicator. It is one of the most important consequences of the universally admitted fact that the 'Lamarckian' theory of inheritance is false.

p. 51 *Another theory, due to Sir Peter Medawar …*

I have been taken to task (not, of course, by Williams himself or even with his knowledge) for attributing this theory of ageing to P. B. Medawar, rather than to G. C. Williams. It is true that many biologists, especially in America, know the theory mainly through Williams's 1957 paper, 'Pleiotropy, Natural Selection, and the Evolution of Senescence'. It is also true that Williams elaborated the theory beyond Medawar's treatment. Nevertheless my own judgement is that Medawar spelled out the essential core of the idea in 1952 in *An Unsolved Problem in Biology* and in 1957 in *The Uniqueness of the Individual*. I should add that I find Williams's development of the theory very helpful, since it makes clear a necessary step in the argument (the importance of 'pleiotropy' or multiple gene effects) which Medawar did not explicitly emphasize. W. D. Hamilton has more recently taken this kind of theory even further in his paper, 'The Moulding of Senescence by Natural Selection'. Incidentally, I have had many interesting letters from doctors but none, I think, commented on my speculations about 'fooling' genes as to the age of the body they are in (pp. 53–4). The idea still doesn't strike me as obviously silly, and if it were right wouldn't it be rather important, medically?

p. 55 *What is the good of sex?*

The problem of what sex is good for is still as tantalizing as ever, despite some thought-provoking books, notably those by M. T. Ghiselin, G. C. Williams, J. Maynard Smith, and G. Bell, and a volume edited by R. Michod and B. Levin. To me, the most exciting new idea is W. D. Hamilton's parasite theory, which has been explained in non-technical language by Jeremy Cherfas and John Gribbin in *The Redundant Male*.

p. 57 *… the surplus DNA is … a parasite, or at best a harmless but useless passenger … (see also p. 237)*

My suggestion that surplus, untranslated DNA might be a self-interested parasite has been taken up and developed by molecular biologists (see papers by Orgel and Crick, and Doolittle and Sapienza) under the catchphrase 'Selfish DNA'. S. J. Gould, in *Hen's Teeth and Horse's Toes*, has made the provocative (to me!) claim that, despite the historical origins of the idea of selfish DNA, 'The theories of selfish genes and selfish DNA could not be more different in the structures of explanation that nurture them.' I find his reasoning wrong but interesting, which, incidentally, he has been kind enough to tell me, is how he usually finds mine. After a preamble on 'reductionism' and 'hierarchy' (which, as usual, I find neither wrong nor interesting), he goes on:

> Dawkins's selfish genes increase in frequency because they have effects on bodies, aiding them in their struggle for existence. Selfish DNA increases in frequency for precisely the